# MIDTERM EXAMINATION
## Fall 2009
## MTH202- Discrete Mathematics

**Question No: 1 ( Marks: 1 ) - Please choose one**

The inverse of given relation R = {(1,1),(1,2),(1,4),(3,4),(4,1)} is

- {(1,1),(2,1),(4,1),(2,3)}
- {(1,1),(1,2),(4,1),( 4,3),(1,4)}
- {(1,1),(2,1),(4,1),(4,3),(1,4)}

**Question No: 2 ( Marks: 1 ) - Please choose one**

Symmetric and antisymmetric are

- Negative of each other
- Both are same
- Not negative of each other

**Question No: 3 ( Marks: 1 ) - Please choose one**

Let A = {a, b, c} and

R = {(a, c), (b, b), (c, a)} be a relation on A. Is R

► Transitive

► Reflexive

► Symmetric

► Transitive and Reflexive

**Question No: 4 ( Marks: 1 ) - Please choose one**

Let A= {1, 2, 3, 4} and R = {(1, 1), (2, 2), (3, 3),(4,4)} then

► R is symmetric.

► R is anti symmetric.

► R is transitive.

► R is reflexive.

► All given options are true

**Question No: 5 ( Marks: 1 ) - Please choose one**

The contrapositive of the conditional statement **p** □ **q** is

► ~p □ ~q

► q □p

► **~q →~p**

► None of these

**Question No: 6 ( Marks: 1 ) - Please choose one**

In Boolean addition 1+1=

► **2**

► **1**

► **0**

**Question No: 7 ( Marks: 1 ) - Please choose one**

The negation of “Today is Friday” is

► Today is Saturday

► **Today is not Friday**

► Today is Thursday

**Question No: 8 ( Marks: 1 ) - Please choose one**

The same element can never appear twice in a set.

► **True**

► False

**Question No: 9 ( Marks: 1 ) - Please choose one**

If f(x)=2x+1, $g(x)=x^2 -1$ then fg(x)=

► $x^2 -1$

► $2x^2 -1$

► $2x^3 -1$

**Question No: 10 ( Marks: 1 ) - Please choose one**

The statement p □q □ ~p □ q □ ~(p □ ~q) describes

► Commutative Law

► Implication Laws

► Exportation Law

► Equivalence

**Question No: 11 ( Marks: 1 ) - Please choose one**

If two sets are not equal, then one must be a subset of the other

► **True**

► False

**Question No: 12 ( Marks: 1 ) - Please choose one**

If f and g are two one-to-one functions then their composition that is gof is

► Not one-to-one

► Onto

► One-to-one

► Onto and one-to-one

**Question No: 13 ( Marks: 1 ) - Please choose one**

If a set contains exactly *m* distinct elements where *m* denotes some non negative integer then the set is .

► Finite

► Infinite

► None of these

**Question No: 14 ( Marks: 1 ) - Please choose one**

If f(4) = 1 and g(1) = 4 then fog(1) =

► 3

► 1

► **4**

**Question No: 15 ( Marks: 1 ) - Please choose one**

If ( $A \cup B$ ) = A, then ( $A \cap B$ ) = B

► True

► False

► Cannot be determined

**Question No: 16 ( Marks: 1 ) - Please choose one**

The total number of elements in a set is called

► Strength

► Cardinality

► Finite

**Question No: 17 ( Marks: 1 ) - Please choose one**

If f(x) = x and g(x)=-2x then (f+g)x =

► 3x

► $2x^2$

► –x

**Question No: 18 ( Marks: 1 ) - Please choose one**

Associative law of Intersection for three sets

► A □ (B □ C) = (A □ B) □ C

► **A ∩ (B ∩ C) = (A ∩ B) ∩ C**

► A □ (B □ C) = (A □ B) □ (A □ B)

► None of these

**Question No: 19 ( Marks: 1 ) - Please choose one**

Which term of the sequence 4,1,-2,… is -77

► **26**

► 27

► 28

**Question No: 20 ( Marks: 1 ) - Please choose one**

If a set A has 5 elements then power set of A. P(S) contains elements. Which are?

- $5^5$
- $2^n$
- $4^5$
- $2^5$

**Question No: 21 ( Marks: 1 ) - Please choose one**

A collection of rules indicating how to form new set objects from those already known to be in the set is called

- Base
- Restriction
- Recursion

**Question No: 22 ( Marks: 1 ) - Please choose one**

$1,10,10^2,10^3,10^4,10^5,10^6,10^7,.....$ is

- Arithmetic series

Arithmetic series

► **Geometric series**

► Arithmetic sequence

► Geometric sequence

**Question No: 23 ( Marks: 1 ) - Please choose one**

$$\sum_{j=1}^{3}(3j-5)=$$

► 4

► **3**

► -2

**Question No: 24 ( Marks: 1 ) - Please choose one**

The product of the positive integers from 1 to n is called

► Multiplication

► n factorial

► **Geometric sequence**

**Question No: 25 ( Marks: 3 )**

Consider the geometric progression 5/7, 5/49, 5/343, …

- Identify r
- Find the nth term of this progression
- If the above progression is to be added up to infinite terms then what could be the sum?

**Answer:-**

$r = ?$

$$r = \frac{a_2}{a_1}$$

$$r = 49/7 \,\big/\, 5/7$$

$$r = \frac{49}{7} \times \frac{7}{5}$$

$$r = \frac{49}{5}$$

nth term=?

$a_1 = 5/7$

$r = 49/5$

$a_n = ?$

*Formula*

$a_n = a_1 r^{n-1}$

$a_n = (5/7)(49/5)^{n-1}$

**Sum=?**

As $n \geq 1$

So, formula is:-

$$S_n = \frac{a(1-r)}{1-r}$$

$$S_n = \frac{5/7(1-49/5)}{1-49/5}$$

$$S_n = \frac{5/7(\frac{5-49}{5})}{\frac{1-49}{5}}$$

$$S_n = \frac{5/7(\frac{-44}{5})}{\frac{-48}{5}} = \frac{\frac{5}{7}\times 44}{48}$$

$$S_n = \frac{5\times 44 11}{7\times 48 12} = \frac{5\times 11}{7\times 12} = \frac{55}{84}$$

$$S_n = \frac{55}{84}$$

**Question No: 26 ( Marks: 5 )**

Determine whether the following statement is tautology or not (use truth table)

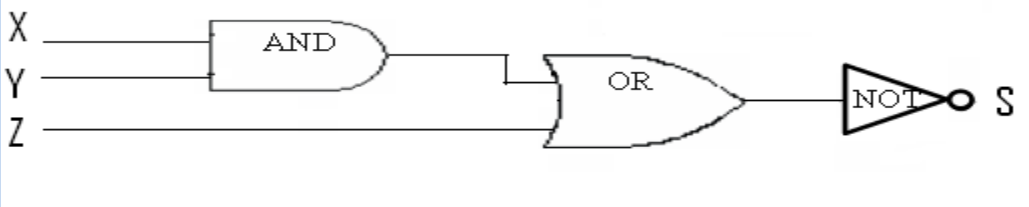
$$\sim q \vee (q \to p) \to \sim p$$

| p | q | ~p | ~q | $q \to p$ | ~q $\vee (q \to p)$ | $\sim q \vee (q \to p) \to \sim p$ |
|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 1 | 1 | 1 | 1 |

Hence, it is not a tautology as there are two false.

**Question No: 27 ( Marks: 10 )**

Write an input/output table for the circuit:




Input signals are:

X = 1

Y = 0

Z = 0

| X | Y | Z | X^Y | (X^Y)∨ Z | ~[(X^Y)∨ Z] |
|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 1 |